SE 1105 - PROGRAMMING AND PROBLEM SOLVING I

Multi-dimensional Array Exercises

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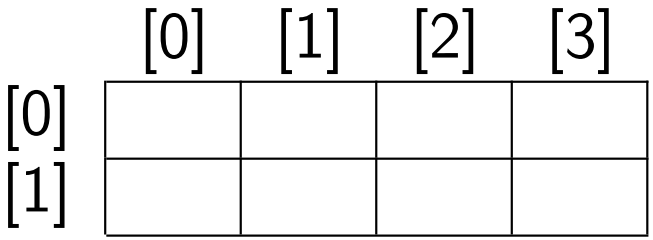
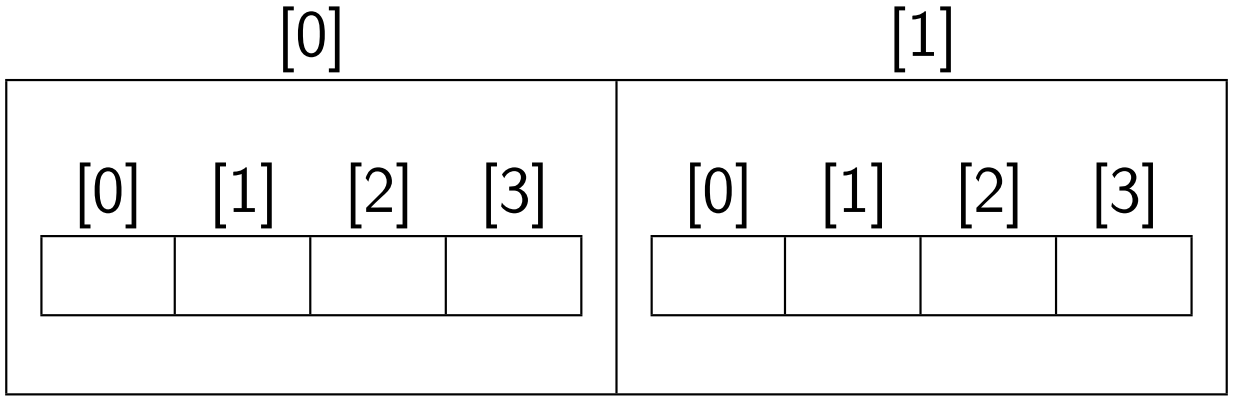
Multi-dimensional Array Review

- Visual Representations of A[2][4]:





- The memory representation A[2][4]

0x000	0x004	0x008	0x00c	0x010	0x014	0x018	0x01c
A[0][0]	A[0][1]	A[0][2]	A[0][3]	A[1][0]	A[1][1]	A[1][2]	A[1][3]

Multi-dimensional Array Review

- Multi-dimensional Arrays can be initialized in the following ways:

```
int A[2][3] = {{1,2,3},{4,5,6}};
```

```
int A[ ][3] = {{1,2,3},{4,5,6}};
```

```
int A[2][3] = {{0}};
```

Multi-dimensional Array Review

- Getting the value of the element at a certain location in a multi-dimensional array:

```
int A[2][3] = {{1,2,3},{4,5,6}};
printf("%d",A[1][2]);
```

- Assigning a value to the element at a certain location in a multi-dimensional array:

```
int A[2][3] = {{1,2,3},{4,5,6}};
A[0][0] = 100;
A[1][0] = A[0][2];
```

Multi-dimensional Array Review

- As we know from one-dimensional arrays, multi-dimensional arrays are also passed to the functions **by reference**.

- It means that when we pass a multi-dimensional array to a function, only the memory address of the very first element of the multi-dimensional array (not the copy of the whole array) is assigned to the related parameter of the function.

0x000	0x004	0x008	0x00c	0x010	0x014	0x018	0x01c
A[0][0]	A[0][1]	A[0][2]	A[0][3]	A[1][0]	A[1][1]	A[1][2]	A[1][3]

Multi-dimensional Array Review

```
#include<stdio.h>
#define M 2
#define N 3

void printMatrix(int A[M][N]){
    for(int i=0; i< M; i++){
        for (int j=0; j< N; j++){
            printf("%d ", A[i][j]);
        }
        printf("\n");
    }
}

void main(){
    int A[M][N] = {{9,-7, 23}, {16, -88, -5}};
    printMatrix(A);
}
```

Passing a Multi-dimensional Array to a Function

```
#include<stdio.h>

void fillMatrix(int m, int n, int A[m][n]){
    for(int i=0; i< m; i++){
        for(int j=0; j< n; j++){
            A[i][j] = i+j;
        }
    }
}

void main(){
    int m = 2, n = 3;
    int A[m][n];
    fillMatrix(m,n,A);
}
```

Exercises

1. Write a C program that calculates (and prints) the grades of 100 students based on the grades given in a 2D array by the formula below. After calculating and printing the grades, your program should also print the entire class's average, min, and max grades.

$$grade = lab * 0.1 + mt * 0.4 + fnl * 0.5$$

2. Write a C program that randomly constructs daily temperature values of 10 weeks for a city and calculates the weekly temperature averages, the hottest and coolest day of these 10 weeks.

Exercises

3. Write a program that simulates a menu for seat selection in a movie theater.

	[0]	[1]	[2]	[3]	[4]	[5]	[6]	[7]	[8]	[9]	[10]	[11]	[12]	[13]	[14]	[15]
[0]	-	X	X	-	-	X	X	X	-	X	-	X	-	X	-	X
[1]	-	X	X	X	-	-	-	X	X	X	-	X	-	-	X	-
[2]	-	-	-	X	X	X	X	X	-	-	X	X	X	-	X	X
[3]	X	X	-	-	-	X	X	-	X	X	-	-	-	X	X	-
[4]	X	X	-	X	X	X	X	X	X	X	X	X	X	X	X	X
[5]	-	-	X	X	-	-	-	-	X	-	-	X	-	X	-	-

Your program should randomly initialize the movie theater seats. Then, it should have a menu for seating choices and payment depending on the student, senior, or adult status. Before and after each ticket purchase, your program should print the seats on the screen in a matrix format.

Exercises

4. Write a program for playing a tic-tac-toe game against the computer.

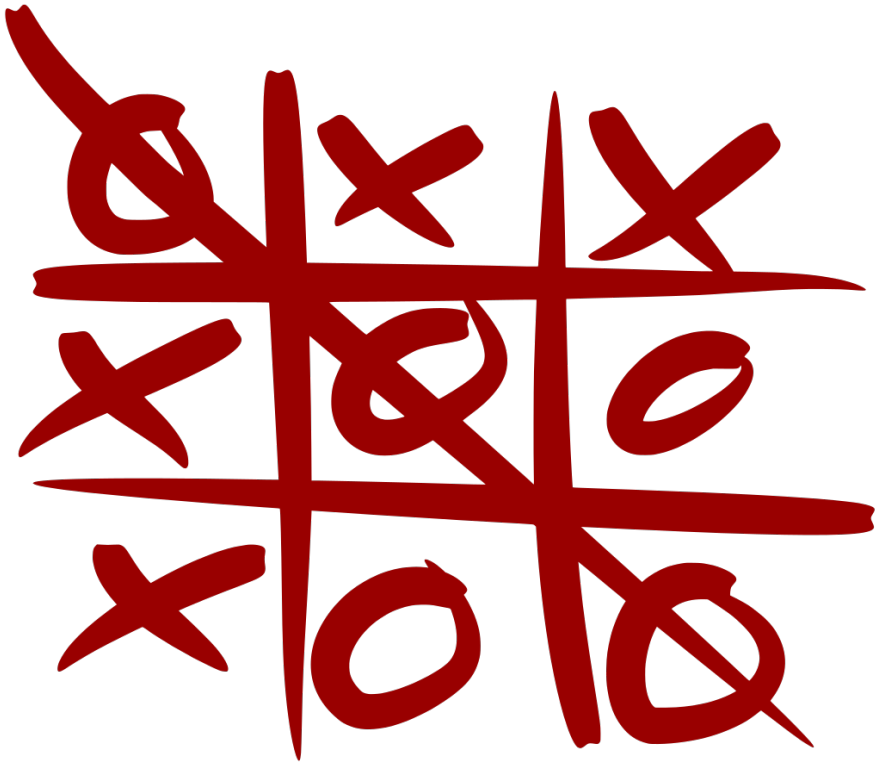


Figure: https://en.wikipedia.org/wiki/Tic-tac-toe